

Internship Report

Title: Job Analytics Portal – Data Visualization Internship

Intern: Payal Kolhe

Organization: Nullclass EdTech Pvt. Ltd.

Duration: October–November 2025

Tools Used: Tableau, Excel, Python (Data Preprocessing)

1. Introduction

This internship at Nullclass EdTech Pvt. Ltd. focused on developing an interactive Job Analytics Portal dashboard to analyze global job market data. The objective was to generate actionable insights through data visualization techniques using Tableau, enhancing understanding of global hiring trends, job categories, and workforce demographics.

2. Background

With the rise of data-driven decision-making, organizations increasingly rely on visual analytics to uncover workforce and hiring trends. This project utilized Tableau to convert raw job data into interactive dashboards, providing clarity across parameters such as company size, qualification, work type, experience, and region.

3. Learning Objectives

- To gain proficiency in Tableau and dashboard creation.
- To understand multi-condition data filtering and visualization logic.
- To develop analytical thinking and data interpretation skills.
- To strengthen real-world problem-solving through practical visualization tasks.

4. Activities and Tasks

The internship comprised six analytical visualization tasks, each designed with specific conditions and objectives. The following describes each task in detail:

Task 1 – Preference vs Work Type (Bar Chart):

A bar chart was created between Preference and Work Type, limited to “Intern.”

- Work Type: Intern
- Latitude: Below 10
- Country: Must not start with A, B, C, or D
- Job Title: Single word with fewer than 10 characters
- Company Size: Below 50,000
- Salary: Above \$9,000
- Experience: Even number
- Job Posting Date: Odd-numbered month

Bars were sorted in descending order by count and configured to display only between 3 PM and 5 PM IST.

Task 2 – Company Size vs Company Name (Scatter Plot):

A scatter plot was drawn between Company Size and Company Name under the following filters:

- Company Size: Less than 50,000
- Job Title: Exactly "Mechanical Engineer"
- Experience: Greater than 5 years
- Country: In Asia, but not starting with "I"
- Salary: Above \$50,000
- Work Type: Full-Time or Part-Time
- Preference: Male
- Job Portal: Idealist
- Company Name: Must have at least two vowels

Chart visibility restricted to 3 PM – 5 PM IST.

Task 3 – Top 10 Companies (Tree Map):

A tree map visualization highlighting the Top 10 companies was created with these conditions:

- Role: Data Engineer
- Job Title: Data Scientist
- Excluded: Asian countries and any country starting with the letter "C"
- Latitude: Below 10
- Preference: Female
- Qualification: Exactly B.Tech
- Job Posting Date: Between 01/01/2023 and 06/01/2023
- Job Portal: LinkedIn
- Company Size: Greater than or equal to 10,000
- Contact Person: Must end with a vowel

Display configured between 3 PM – 5 PM IST.

Task 4 – Qualification Drilldown Map (Map with Click):

A map visualization showing qualification-based postings in Africa was implemented.

- Qualification: B.Tech, M.Tech, or PhD
- Work Type: Full-Time
- Country: Only African nations
- Job Title: Must start with "D"
- Preference: Male
- Company Size: Greater than 80,000
- Contact Person: Must start with "A"
- Job Portal: Indeed
- Salary Range: Above \$20,000

Each plotted point opened a drilldown map showing precise location coordinates. Display limited to 3 PM – 6 PM IST.

Task 5 – India vs Germany Comparison (Stacked Bar Chart):

A stacked bar chart compared job postings between India (Orange) and Germany (Green).

- Qualification: B.Tech
- Work Type: Full-Time
- Experience: Greater than 2 years
- Job Title: Data Scientist, Art Teacher, or Aerospace Engineer
- Salary: Above \$10,000
- Job Portal: Indeed
- Preference: Female
- Job Posting Date: Before 08/01/2023
- Location: Must not be blank
- Company Name: Must have more than 8 characters

Visible between 3 PM and 5 PM IST.

Task 6 – Work Type Salary Distribution (Box Plot):

A box-and-whisker plot visualizing salary range by Work Type = “Intern.”

- Latitude: Below 10
- Country: Must not start with A, B, C, or D
- Job Title: Single word with fewer than 10 characters
- Company Size: Less than 50,000
- Salary: Above \$8,000
- Experience: Even number
- Job Posting Date: Between 2021–2023
- Contact Person: Must contain at least one “e”

Chart visibility restricted between 3 PM and 5 PM IST.

5. Skills and Competencies Developed

- Technical Skills: Tableau dashboards, data preprocessing, map integration, and conditional logic application.
- Analytical Skills: Multi-criteria filtering, insight generation, and data interpretation.
- Professional Skills: Problem-solving, documentation, and task management.

6. Feedback and Evidence

Each visualization was reviewed internally for accuracy and presentation quality. Screenshots served as verification, confirming correct application of all conditions. Feedback appreciated the clarity, consistency, and interactive nature of the dashboards.

7. Challenges and Solutions

Challenge: Managing multi-condition filters → **Solution:** Sequential validation and use of logical hierarchies.

Challenge: Ensuring visual uniformity → **Solution:** Adopted consistent color palette and layout.

Challenge: Time-restricted display implementation → **Solution:** Applied conditional dashboard logic in Tableau.

Challenge: Handling missing data fields → **Solution:** Pre-cleaned data in Excel before importing to Tableau.

8. Outcomes and Impact

The Job Analytics Portal successfully transformed raw job data into interactive visual insights. It provided clarity on hiring trends, salary distribution, and global workforce analytics, enhancing both analytical understanding and visualization expertise.

9. Conclusion

This internship at Nullclass EdTech Pvt. Ltd. offered an excellent opportunity to apply data analytics concepts to real-world visualization tasks. By completing six complex dashboards, strong technical and analytical skills were developed, reinforcing readiness for professional data analytics roles.